PRIVACY · REAL-TIME · RESILIENT

ROI Privacy-Preserving Edge Streaming

A Resilient Surveillance System for Unreliable Networks

ICT Application Technology — Final Project

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The Surveillance Paradox

THE PROBLEM

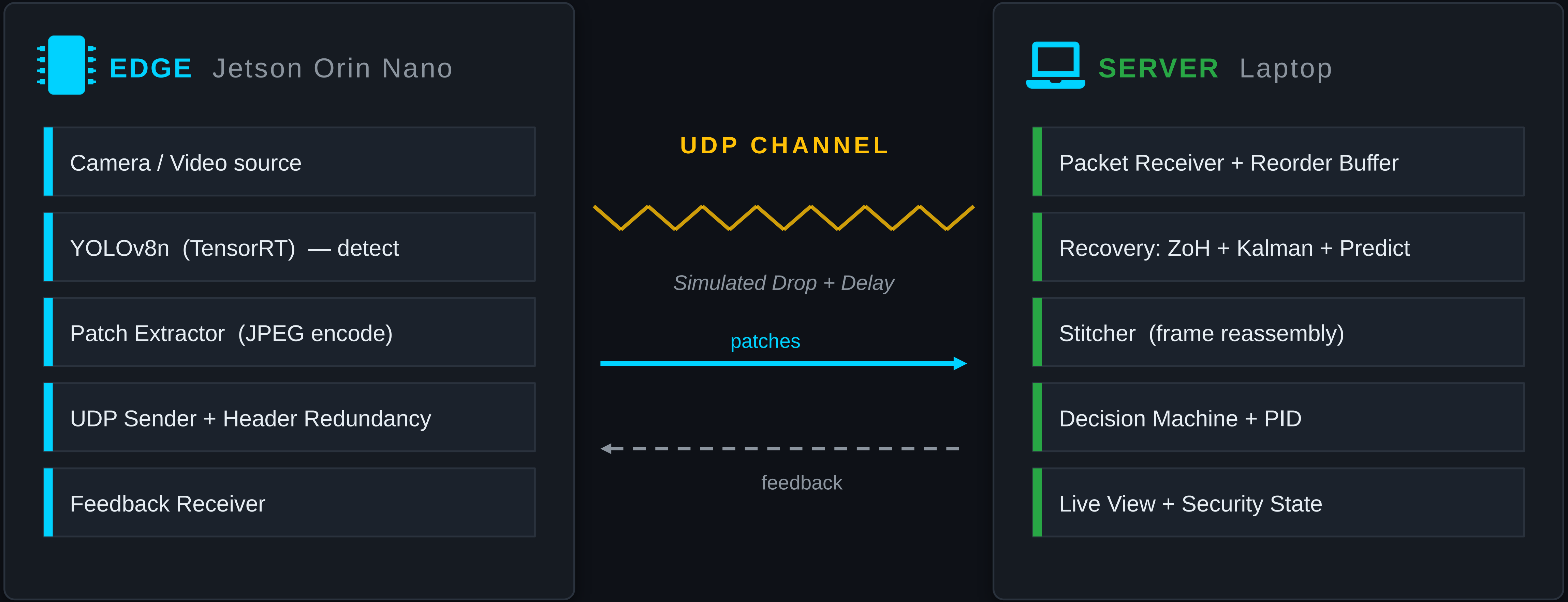
- **Full video transmission raises** privacy concerns
- **Networks are unreliable:** packets drop, latency spikes
- **Bandwidth constraints in** distributed surveillance

OUR SOLUTION

- **Transmit only ROI (people)** — background discarded at edge
- **Real-time recovery** from packet loss
- **Adaptive quality control** via closed-loop PID

System Architecture

Two Logically Separated Nodes



Privacy: background discarded at edge — never transmitted

Adaptive: closed-loop quality control via PID feedback

Four Layers of Recovery

Frame Header Redundancy
01

Each frame header sent 5×.
At 50% packet drop, total-loss probability = 3%. Frames almost never lost.

Zero-order Hold Recovery
02

When a patch fails to arrive, reuse the previous frame's patch image. The person doesn't disappear.

Kalman Compensation
03

Estimates each person's velocity and translates the recovered patch to the predicted position. Eliminates misalignment.

Predict-only Mode
04

When detection itself fails, Kalman generates virtual patches at expected positions. Tracking continues.

Adaptive Control Loop

PID Controller adjusts JPEG quality (30–95) from reception ratio

3-State Machine Normal / Warning / Emergency

Live Demo

Real-time system response to network degradation

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Privacy by Design

Only ROI patches transmitted — background discarded at the edge

Adaptive by Necessity

PID controller closes the loop on JPEG quality from reception ratio

Resilient by Engineering

Four layers of recovery: from frame headers to predict-only mode

Thank you.

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